

XDJ100SX

Open Source Standalone Deck



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INDEX

Introduction.....	2
Phase 0: disassembling.....	4
Phase 1: soldering.....	6
Phase 2: Firmware.....	12
Phase 3: Midi Mapping.....	16
Phase 4: Skin.....	21
Phase 5: Linux System.....	24
Phase 6: Physical skin.....	28
Phase 7: Assemble everything.....	33
Phase 8: Improvements.....	35
8.1 Power on/off.....	35
8.2 USB Eject.....	38
8.3 USB Network Share.....	42
Using the image.....	43

Introduction

I wanted to take an old DJ deck to convert into a modern one and to be standalone.

What standalone stands for?

Any DJ deck has a computer inside, usually under a Linux operating system which is 100% optimized for that device + a software which is the one that can read, play and manipulate tracks with different features. The software also has a dedicated look/skin so perhaps for a standard user this does not look like a computer / software, but it certainly is.

In conclusion, a standalone deck is a controller that controls a dedicated software that is running in a mini computer where everything fits together in a box.

Goal

Recently AlphaTheta dropped their CDJ-3000X, the successor of the 3000 which apart from a bigger screen and some additional hardware additions, almost all features could be upgradable with a firmware / software update. This is a way of business, as many other brands like Apple's iPhone which gets renewed every year, and makes the previous release obsolete, or scheduled obsolescence.

For both standard and professional users, it's a way of having to spend lots of money to invest in something that won't be upgradeable and will be obsolete in less than 5 years, feeling also that it should be thrown away or to sell to the second hand market.

That's one of the reasons I wanted to make a fully functional, modern 20 year old device using open source software and make it friendly for any DJ user used to CDJ decks.

Be in mind I am not the first to do it. There are some other cool projects and modifications made by some other users. Unfortunately as far as I could find, they were not 100% open, meaning they depended on some licensed tools.

And so the second reason:

I wanted to make this project 100% open, not only in terms of sharing the code, but also using 100% open source software, and open source OS.

This project can be replicated to any other device. Codes should slightly change but the basis is the same.

How to make it work?

We just need to use an existing deck, and make all buttons to send MIDI through a firmware. The firmware will be Arduino based. This is the first open source part of this project.

But why MIDI? The goal is to have a standalone deck with lots of features that any other deck can do. There is already an existing DJ application which is 100% open source and works in Mac, Windows and Linux with different system architectures: Mixxx. This app has an amazing community of developers, djs and enthusiasts that make that anyone can DJ with any system, DJ setup. It's open and we will be able to customize and adapt to our project. This app can send and receive MIDI so it's the easiest way to communicate with it.

How then? Using a computer that we will attach inside the device. This computer will run Linux which is also an open source operating system. There are some options, I decided to go for the established Raspberry Pi Foundation that contributes to the Linux and open source community.

The CDJ-100S:**Requirements used for each deck:**

- Raspberry Pi 3B+
- SD Card 64 GB Class A2 U3
- Teensy LC
- Power Supply 220V AC / 5V DC 3A
- Capacitive Touchscreen 5" DSI Port
- PCB Cables
- Stereo audio cable
- Rotary Encoder with button
- USB connector Type A
- Terminal connectors

Tools needed:

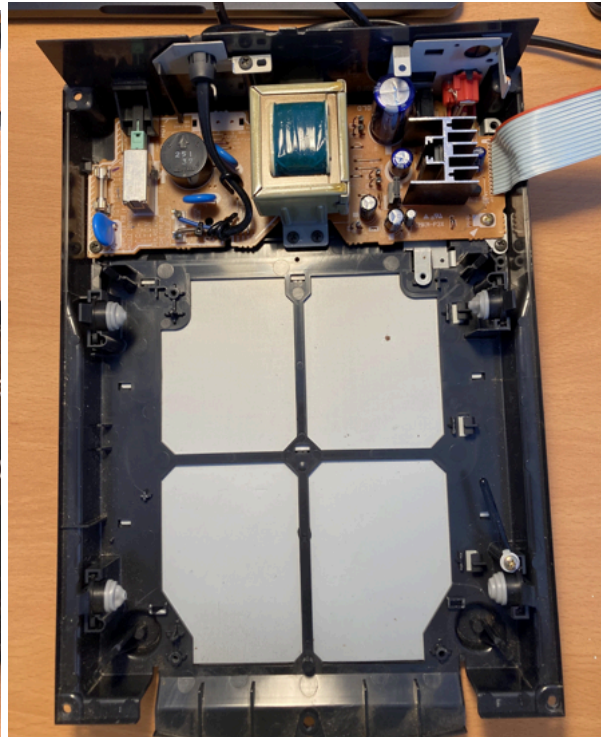
- Multimeter
- Soldering Station
- Screwdrivers
- Cutting Plier
- Arduino IDE app
- Visual Studio Code
- Midi Monitor

Phase 0: disassembling

Disassemble all the CDJ and remove all PCB and not desired parts: CD unit, main PCB and power supply. Some cables are attached to the front panel, so we also need to disconnect all of them.

What we want to keep:

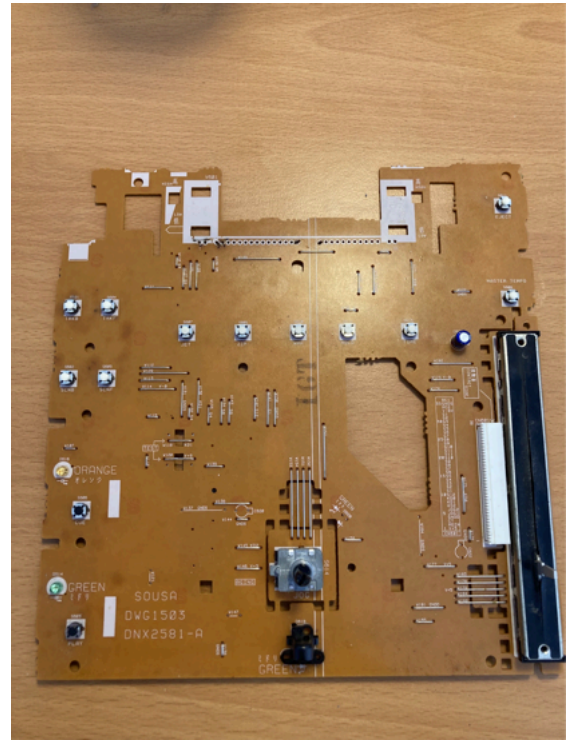
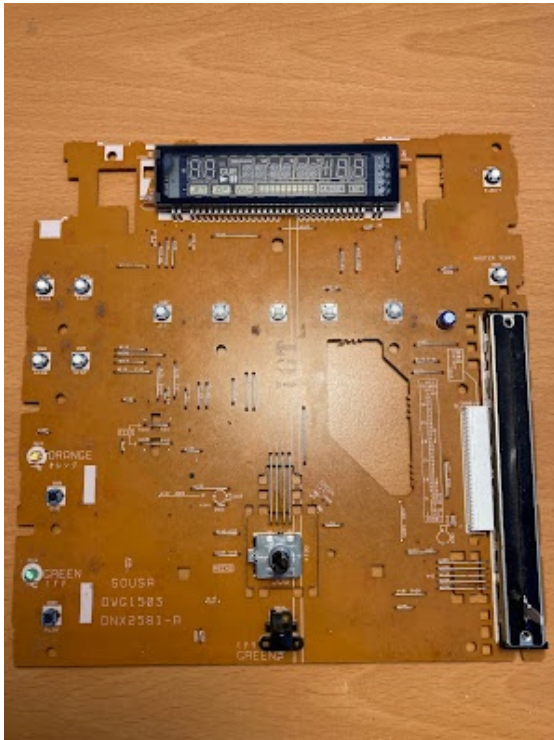
- Power switch (on/off): it's attached to the board, so we can just break/cut the board to keep the part where the switch is soldered.
- RCA audio connector: it's also soldered on the board. We need to desolder it.
- Main power cable.



First figure: after removing the front panel.

Right figure: after removing the main PCB.

Front panel: we need to remove the old screen desoldering all pins (it may take a while):



Phase 1: soldering

The first thing we have to do is to check all original buttons and leds (in addition to the ones we want to add). This is important so we must know how many connections we have before getting the Teensy/Arduino PCB.

The CDJ100S has:

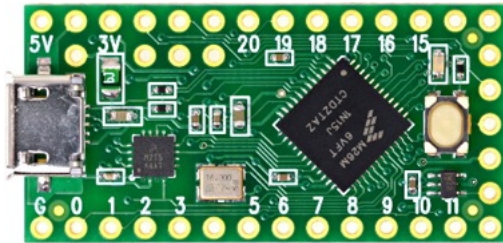
Name	Type	In/out	Number of pins	Analog / Digital?
Eject	Button	IN	1	Digital
Track Search Previous	Button	IN	1	Digital
Track Search Next	Button	IN	1	Digital
Search Back	Button	IN	1	Digital
Search Forward	Button	IN	1	Digital
Cue	Button	IN	1	Digital
Play/Pause	Button	IN	1	Digital
Jet	Button	IN	1	Digital
Zip	Button	IN	1	Digital
Wah	Button	IN	1	Digital
Hold	Button	IN	1	Digital
Time mode / Auto Cue	Button	IN	1	Digital
Master Tempo	Button	IN	1	Digital
Pitch	Slider	IN	1	Analog
Jog Wheel	Encoder	IN	2	Digital
Cue Led	Led	OUT	1	Digital
Play Led	Led	OUT	1	Digital
Internal Led	Led	OUT	1	Digital
CD Led	Led	OUT	1	Digital
Browse (addition)	Encoder	IN	2	Digital
Load (addition)	Button	IN	1	Digital
TOTAL			23	

So the CDJ100S has a total of 20 pins. It does not have any encoder to browse so we will add one encoder with a button and we will be able to navigate through the library and load a track. This adds 3 additional pins, so we need a board with at least 23 pins.

The pitch slider and encoders need a tension of 3,3 V to work. We must know that our board also has a Vout of 3,3 V. Also, the pitch slider has to be read as analog so we need analog input.

Now that we know that we need 23 pins we check in the market what's the best option. To work with Mixxx we will use the MIDI protocol, so we have to be sure that our board is compatible with MIDI libraries.

I've found the Teensy LC has 26 pins, it does have analog/digital and 3,3V Vout too.

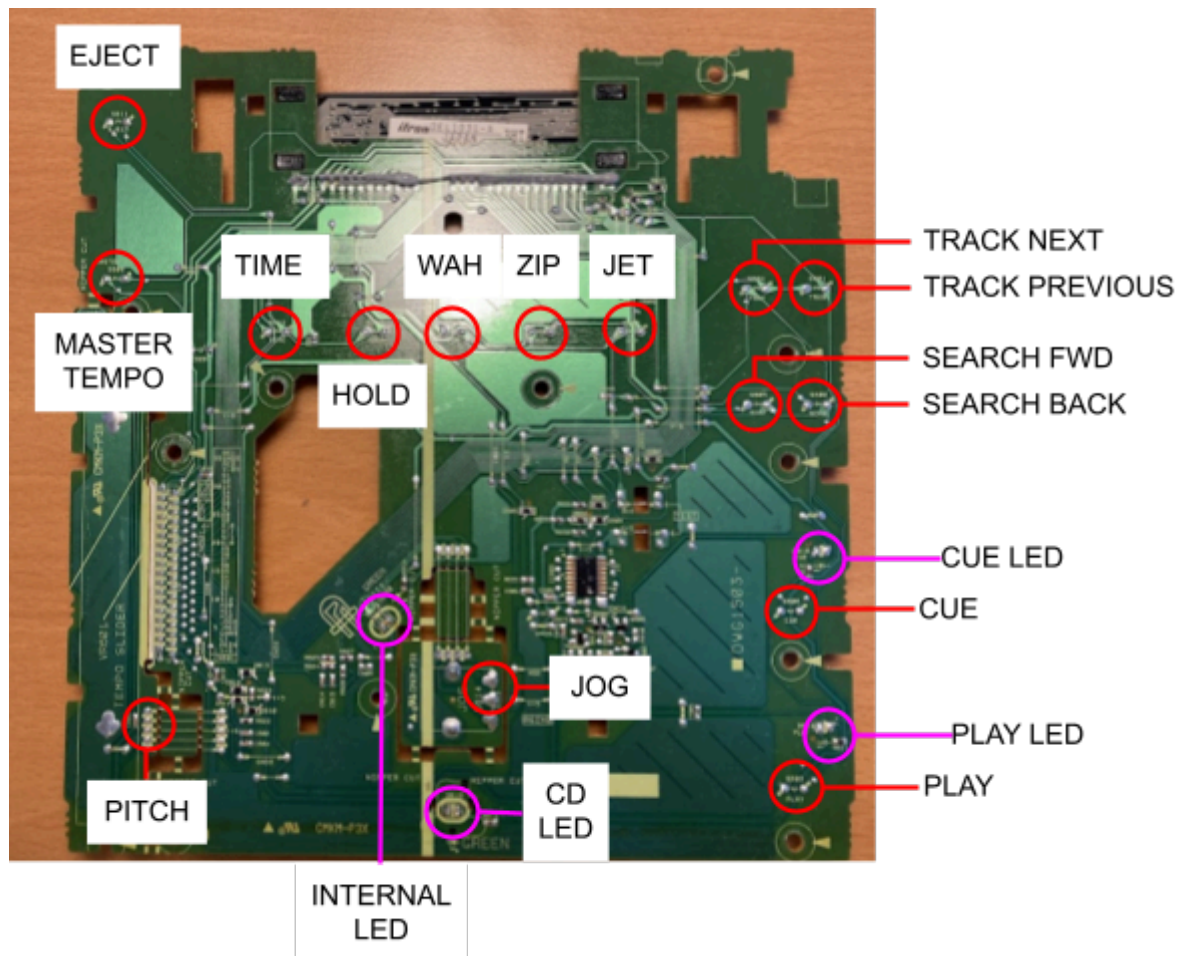


Note: any board which is compatible with Arduino and follows the above requirements with the desired Pins will work.

So we need to solder all the pins to our board. This is what I did for the Teensy LC:

Name	Type	In/out	PIN Number
Eject	Button	IN	0
Track Search Previous	Button	IN	1
Track Search Next	Button	IN	2
Search Back	Button	IN	3
Search Forward	Button	IN	4
Cue	Button	IN	5
Play/Pause	Button	IN	6
Jet	Button	IN	7
Zip	Button	IN	8
Wah	Button	IN	9
Hold	Button	IN	10
Time mode / Auto Cue	Button	IN	11
Master Tempo	Button	IN	12
Pitch	Slider	IN	14 + 3,3V
Jog Wheel	Encoder	IN	15 & 20
Cue Led	Led	OUT	16
Play Led	Led	OUT	17
Internal Led	Led	OUT	18
CD Led	Led	OUT	19
Browse (addition)	Encoder	IN	21 & 22 +3,3V
Load (addition)	Button	IN	23

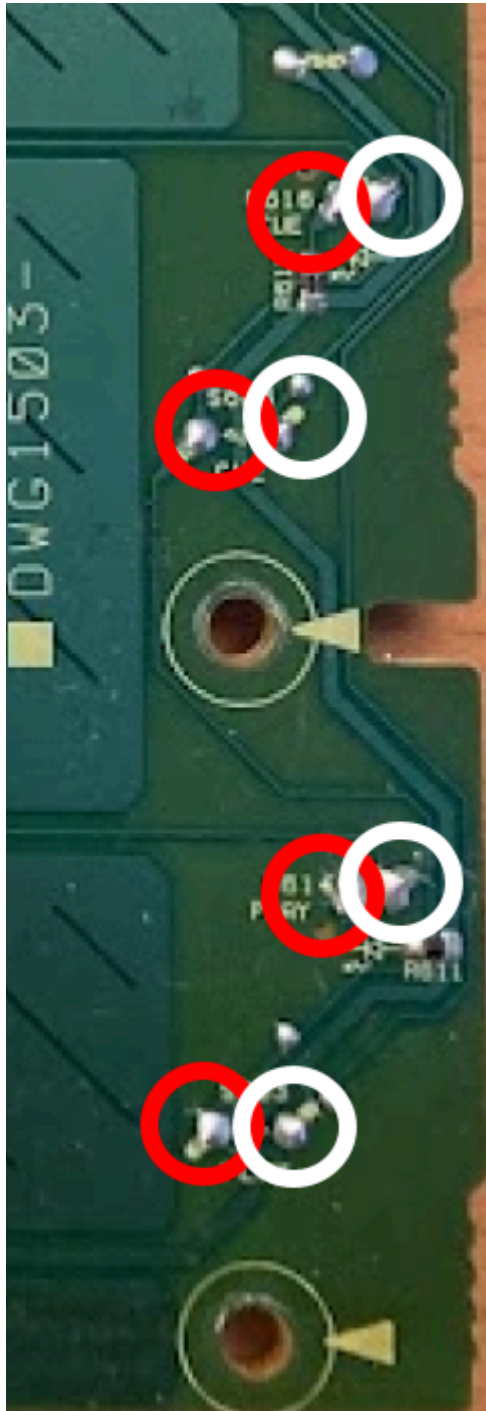
Now we need to know where we have to solder the board, so we identify where are them, using the multimeter:



Note: each pin must be connected to the ground too, this means we need the pin and a second wire to be grounded to every pin.

If you are not used to this, basically you have to make a new ground line and this ground line will be soldered to the arduino board to the ground pin.

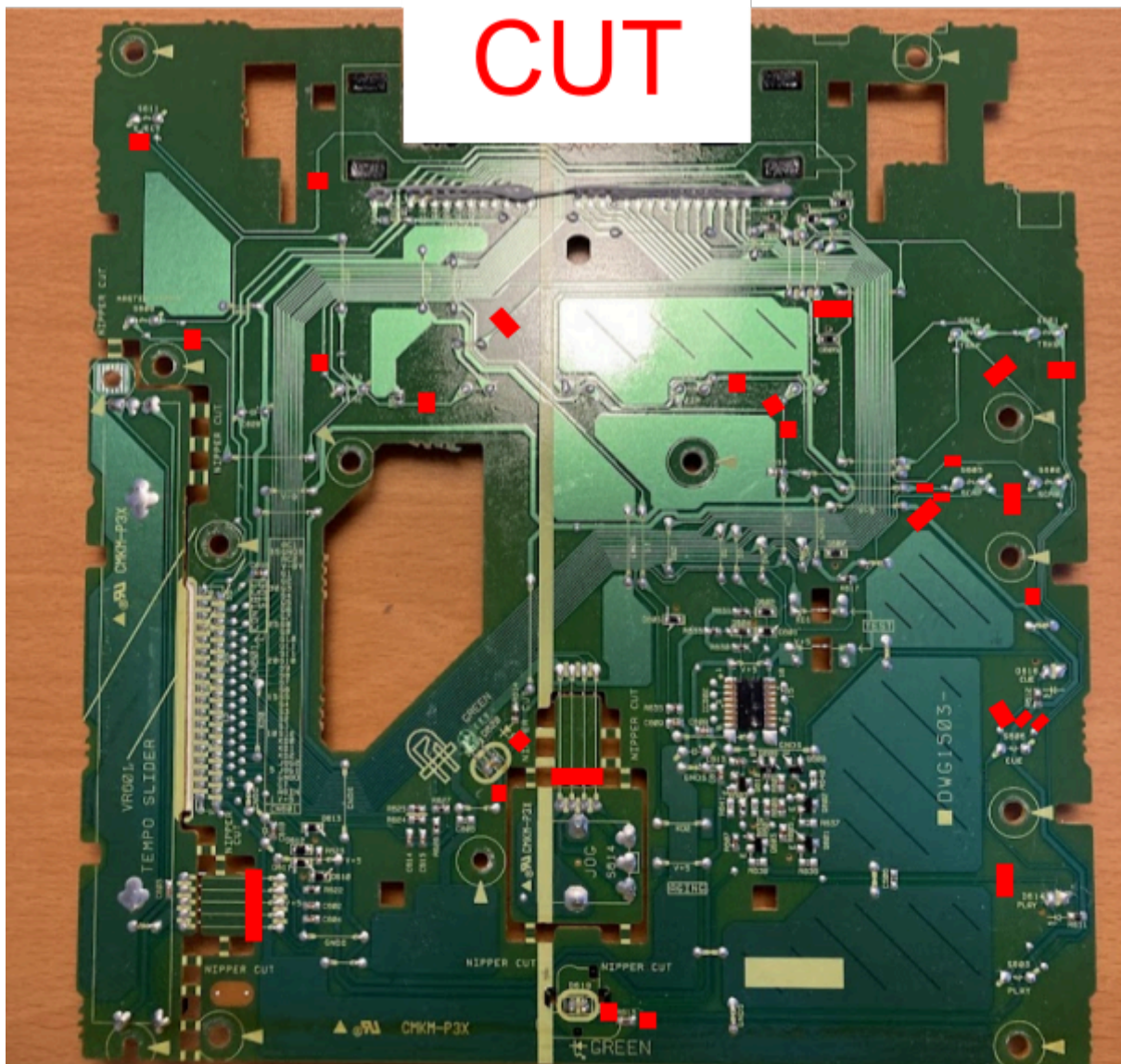
In this figure, Play and Cue buttons and leds:



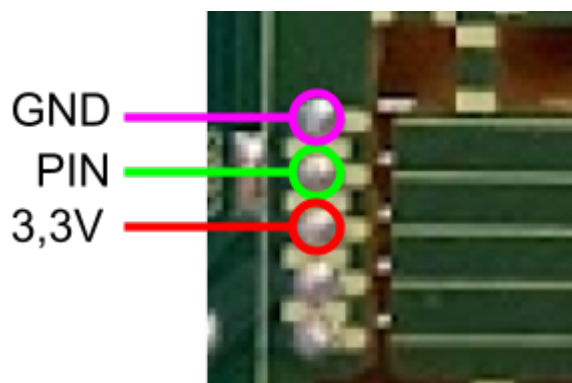
RED = GROUND
WHITE = PIN

So with every PIN you must put its ground to the ground line. Polarity only matters for the LEDs here.

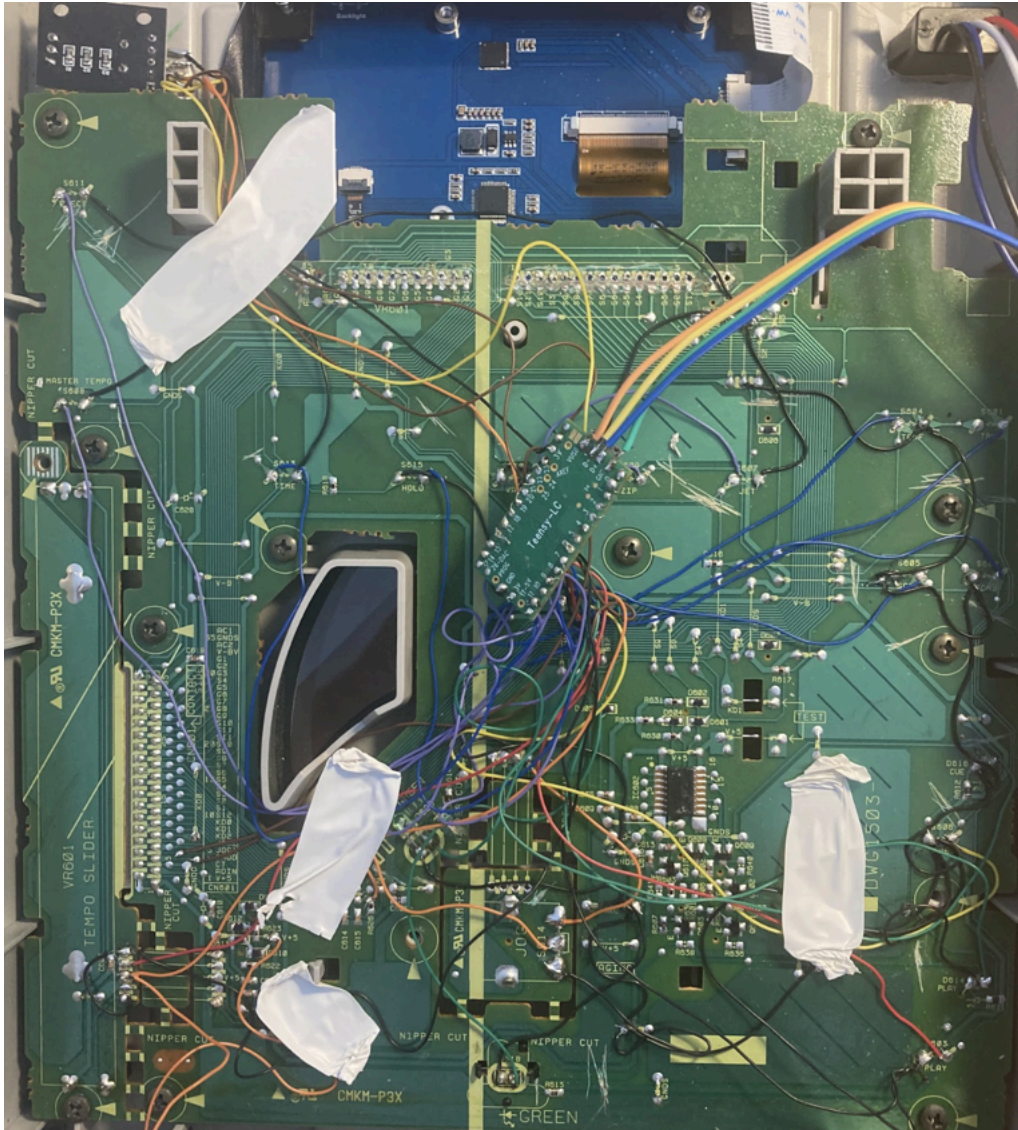
The CDJ100S has also some shorted pins that will make some of our new design not to be working, so we need to carefully cut them in order to make the new ground line. A multimeter is your friend here. We need to cut these:



The Pitch slider has several solder points, after checking in the CDJ service manual and using a multimeter I found these are the ones we have to use:

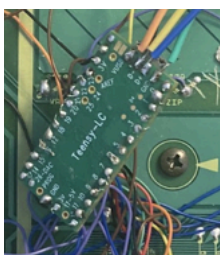


So with all done, it should look something like this:



Note the Teensy board is floating in the center. You can also see the new browser encoder on the top left.

Apart from that, I later soldered the 4 USB pins instead of using an USB connector.



Phase 2: Firmware

Our friend here is the Arduino IDE app. You may ignore this part if you want to upload my code directly.

Implementation

So basically every time we do something, Arduino should send a MIDI message.

Type of MIDI messages:

Note On / Note Off : it can be called from a button or switch.

Control Change (CC): sends different values according to the position of the physical component. This can be called from knobs and sliders (jog, browse and pitch).

Also, the default led is considered a switch, so it has two states: on & off so it receives a Note On / Off.

This sounds easy, and I was not used to it until I saw that there were bounces.

In the real world we have bounces which means that any button, knob, etc, can have interference or physical bounce so we need to add a delay in every Pin so we can avoid a repeat touch. I've found there is already an implementation in the [Bounce library](#) that fixes this so we can avoid writing a new code. We only need to include it in our code and call it for every button. In my code, I added 50 ms of delay. This is maybe not the best for the CUE buttons if we want to make fast rhythms, so you may want to change to a lower value.

```
C/C++
#include <Bounce.h>
Bounce cue_boto = Bounce (cue_pin, 50);
```

For the encoders, we also need to implement if we are going forward or backwards. There is also the [Encoder library](#) which keeps track of the direction.

```
C/C++
#include <Encoder.h>
Encoder jog(jogA_pin, jogB_pin);
long lastPosition_jog = 0;
```

For the Pitch Slider, I've found several ways to make it work, but the [ResponsiveAnalogRead library](#) worked the best.

```
C/C++
#include <ResponsiveAnalogRead.h>
ResponsiveAnalogRead analog(pitch_pin, true);
```

Leds

This device has 4 leds: Cue, Play, Internal and CD. The first two will turn on as expected. The internal and CD are the ones we have to decide what to do. So I decided that the internal will blink to the track beat as Mixxx can send Beat through MIDI. For the CD led, it is going to blink when the track is ending. Mixxx can also send a 'true' value when the track is about to finish.

So the code is basically a *digitalWrite(pin, HIGH)* when the arduino receives the MIDI note. In the example below, I set '61' to be the MIDI note that Mixxx sent while the track is playing or while it is paused.

While playing: Mixxx sends the note 61 with a velocity=127

While paused: Mixxx sends the note 61 with a constant loop velocity of 127 and 0 (blinking).

```
C/C++
const int PLAY_NOTE_INDICATOR = 61;
[...]
void handleNoteOn(byte channel, byte note, byte velocity) {

if (note == PLAY_NOTE_INDICATOR) {
  if (velocity > 0) {
    digitalWrite(ledPlay, HIGH); //led enabled
  }
  else {
    digitalWrite(ledPlay, LOW); //led disabled
  }
}
```

For the CD led, I set the note to be 64. Then Mixxx sends 'true' if the track is ending and 'false' if it's not. I needed to make a loop to make a blink.

```
C/C++
#include <elapsedMillis.h>
const int LEDCD_NOTE_INDICATOR = 64;
bool blink = false;
unsigned long previousTime = 0;

if (note == LEDCD_NOTE_INDICATOR) {
  blink = (velocity > 0);
}
}

if (blink){
  if (millis () - previousTime >= 1000){
    previousTime = millis();
  }
}
```

```
        digitalWrite(ledCd, !digitalRead(ledCd));
    }
}
else{
    digitalWrite(ledCd, LOW);
}
```

Main setup

For every pin we need to set if it's an INPUT and OUTPUT:

```
C/C++
pinMode(eject_pin, INPUT_PULLUP);
[...]
pinMode(ledCue, OUTPUT);
[...]
```

We also need to initialize our MIDI receive functions and both encoders.

```
C/C++
usbMIDI.setHandleNoteOn(handleNoteOn);
usbMIDI.setHandleNoteOff(handleNoteOff);
jog.write(0);
browse.write(0);
lastPosition_browse = 0;
```

Void Loop

We need to update every component and define what they do. For the buttons, I used the *'fallingEdge'* and *'risingEdge'* to read the state (pressed / unpressed):

```
C/C++
eject_boto.update();
if(play_boto.fallingEdge()){
    usbMIDI.sendNoteOn(60, 127, channel);
}
if(play_boto.risingEdge()){
    usbMIDI.sendNoteOff(60, 0, channel);
}
```

Midi values

These are the MIDI values I implemented. You can use any other. The translated hex note is important for the mapping later.

Name	Midi Channel	Type	MIDI Note	Hex Note
Eject	1	Note On	63	0x3F
Track Search Previous	1	Note On	64	0x40
Track Search Next	1	Note On	65	0x41
Search Back	1	Note On	66	0x42
Search Forward	1	Note On	67	0x43
Cue	1	Note On	61	0x3D
Play/Pause	1	Note On	60	0x3C
Jet	1	Note On	68	0x44
Zip	1	Note On	69	0x45
Wah	1	Note On	70	0x46
Hold	1	Note On	71	0x47
Time mode / Auto Cue	1	Note On	72	0x48
Master Tempo	1	Note On	62	0x3E
Pitch	3	CC	7	0x07
Jog Wheel	2	CC	20	0x14
Cue Led	1	Note On	62	0x3E
Play Led	1	Note On	61	0x3D
Internal Led	1	Note On	63	0x3F
CD Led	1	Note On	64	0x40
Browse (addition)	3	Note On	70 & 71	0x46 & 0x47
Load (addition)	1	Note On	60	0x3C

Apart from that, we must know the hexadecimal translations for the message type. The message also contains the MIDI channel:

Note On: 0x90, 0x91, 0x92, ... (90 for MIDI channel 1, 91 for channel 2, 92 for channel 3)

Note Off: 0x80, 0x81, 0x82, ...

CC: 0xB0, 0xB1, 0xB2, ... (B0 for MIDI channel 1, B1 for channel 2, B2 for channel 3)

Phase 3: Midi Mapping

Mixxx allows us to make custom mappings. These mappings are made using XML code which is similar to HTML. It's basically a tree where you can define what MIDI message makes software action. This goes in conjunction with an optional Javascript file, which allows us to make complex mappings, variables or making a button to make more than one action.

Now, we should define what we want to do with every button. For this model, these are the actions I decided to map:

Name	Midi Channel	Type	Hex Note	Action	Second action (shift)
Eject	1	Note On	0x3F	Browse Back	-
Track Search Previous	1	Note On	0x40	Loop Half	-
Track Search Next	1	Note On	0x41	Loop Double	-
Search Back	1	Note On	0x42	Search Back	Faster Search back
Search Forward	1	Note On	0x43	Search Forward	Faster Search forward
Cue	1	Note On	0x3D	Cue	Go to start
Play/Pause	1	Note On	0x3C	Play/Pause	-
Jet	1	Note On	0x44	Button 1	Depends on button mode
Zip	1	Note On	0x45	Button 2	Depends on button mode
Wah	1	Note On	0x46	Button 3	Depends on button mode
Hold	1	Note On	0x47	Shift Button	-
Time mode / Auto Cue	1	Note On	0x48	Button Mode	-
Master Tempo	1	Note On	0x3E	Master Tempo	Tempo Range
Pitch	3	CC	0x07	Pitch	-
Jog Wheel	2	CC	0x14	Jog	-
Cue Led	1	Note On	0x3E	Cue Indicator	-
Play Led	1	Note On	0x3D	Play Indicator	-
Internal Led	1	Note On	0x3F	BPM Led	-
CD Led	1	Note On	0x40	End warning	-
Browse (addition)	3	Note On	0x46 & 0x47	Browse Up/Down / Waveform Zoom in/out	-
Load (addition)	1	Note On	0x3C	Load Track / Load playlist	-

I wanted to have multiple modes (layers), so the three original *Jet*, *Zip* & *Wah* can make different actions.

I decided to have 6 layers / button modes:

1st MODE: HOT CUE: A, B, C

2nd MODE: HOT CUE: D, E, F

3rd MODE: HOT CUE: G, H

4th MODE: LOOP ROLL: $\frac{1}{4}$, $\frac{1}{2}$, 1

5th MODE: BEATJUMP: LEFT, RIGHT, DEFINE LENGTH

6th MODE: KEYSHIFT: SEMITONE DOWN, SEMITONE UP, RESET

To implement the different modes, we can create a variable, but the issue here is that we should have visual feedback on the skin and after doing some research I did not find a way to have this visual variable in the skin, so I managed to map every mode to a non used parameter in the mixer. So these are the EQ Kill, so:

1st MODE: DECK2 LOW KILL

2nd MODE: DECK2 MID KILL

3rd MODE: DECK2 HI KILL

4th MODE: DECK3 LOW KILL

5th MODE: DECK3 MID KILL

6th MODE: DECK3 HI KILL

Of course we can even add the DECK4 (and DECK1 as we are not using the internal mixer here), but I kept it to 6 modes.

The Javascript file has to be included in the XML:

XML

```
<scriptfiles>
  <file filename="XDJ100SX.js" functionprefix="XDJ100SX"/>
</scriptfiles>
```

The play button is that simple:

```
XML
<control>
  <group>[Channel1]</group>
  <key>play</key>
  <description>PLAY</description>
  <status>0x90</status>
  <midino>0x3C</midino>
  <options>
    <normal/>
  </options>
</control>
```

So the main parameters are:

group: Deck number

key: mixxxx action

status: MIDI message type

midino: MIDI message note

But let's see now the CUE button, which I want also that a Shift button + CUE make the player go to the beginning of the track (like any other controller):

```
XML
<control>
  <group>[Channel1]</group>
  <key>XDJ100SX.cue</key>
  <description>CUE</description>
  <status>0x90</status>
  <midino>0x3D</midino>
  <options>
    <script-binding/>
  </options>
</control>
```

So:

key: name_of_the_javascript_object.name_of_the_function

options: we have to add <script-binding/> to let it know that this 'key' is in the javascript file

It calls the function *XDJ100SX.cue* that I declared in the Javascript file.

Javascript

I took some variables from some other controllers so I could easy implement them into my code.

Additionally, the toggle button mode that toggles between the different EQ Kills:

JavaScript

/Button Mode

```
XDJ100SX.buttonMode = function (channel, control, value, status, group) {

    if (value > 0) {
        // Go to next mode
        XDJ100SX.currentMode = (XDJ100SX.currentMode + 1) % 6;

        // Set all to 0
        engine.setValue("[Channel2]", "filterLowKill", 0);
        engine.setValue("[Channel2]", "filterMidKill", 0);
        engine.setValue("[Channel2]", "filterHighKill", 0);
        engine.setValue("[Channel3]", "filterLowKill", 0);
        engine.setValue("[Channel3]", "filterMidKill", 0);
        engine.setValue("[Channel3]", "filterHighKill", 0);

        // Enable the current one
        if (XDJ100SX.currentMode === 0) {
            engine.setValue("[Channel2]", "filterLowKill", 1); //Mode 1
        } else if (XDJ100SX.currentMode === 1) {
            engine.setValue("[Channel2]", "filterMidKill", 1); //Mode 2
        } else if (XDJ100SX.currentMode === 2){
            engine.setValue("[Channel2]", "filterHighKill", 1); //Mode 3
        }
        else if (XDJ100SX.currentMode === 3){
            engine.setValue("[Channel3]", "filterLowKill", 1); //Mode 4
        }
        else if (XDJ100SX.currentMode === 4){
            engine.setValue("[Channel3]", "filterMidKill", 1); //Mode 5
        }
        else{
            engine.setValue("[Channel3]", "filterHighKill", 1); //Mode 6
        }
    }
};
```

And then an example for the 3 buttons in the first mode (currentMode = 0):

JavaScript

```

XDJ100SX.button = function(buttonNumber){
    return function (channel, control, value, status, group){
        if(value === 127){

            //Hot Cue A, B, C
            if(XDJ100SX.currentMode === 0){
                if(XDJ100SX.shiftPressed){
                    engine.setValue(group, "hotcue_" + buttonNumber + "_clear",
1);
                }
                else{
                    engine.setValue(group, "hotcue_" + buttonNumber +
"_activate", 1);
                }
            }
        }
    }
}

```

As I added a browse encoder, I mapped the button to make different things:

When we see the waveform -> zoom in / zoom out

When we see the library -> scroll up / down

In Mixxx we can also switch between the *Tabs* declared in the Skin, so:

JavaScript

```

XDJ100SX.browseDown = function(channel, control, value, status, group) {
    if (value === 127) {
        // llegim el control actual
        var currentTab = engine.getValue("[Tab]", "current");
        if (currentTab === 0) { // 0 = overview
            engine.setValue("[Channel1]", "waveform_zoom_down", 1); // 1 =
library
        }
        // Enviem moviment al browser
        engine.setValue("[Library]", "MoveDown", 1);
    }
};

XDJ100SX.browseUp = function(channel, control, value, status, group) {
    if (value === 127) {
        var currentTab = engine.getValue("[Tab]", "current");
        if (currentTab === 0) {
            engine.setValue("[Channel1]", "waveform_zoom_up", 1);
        } engine.setValue("[Library]", "MoveUp", 1);
    }
}

```


Phase 4: Skin

The goal here is not only to integrate the Skin to the small 5 inches touchscreen but also to make it look like a standard Pioneer/AlphaTheta deck as most of the DJs are used to this layout.

After doing some research, I found the user [Timewasternl](#) showed a video where he contributed at creating a custom Mixxx skin that looked similar to a 2 Pioneer deck so he converted a DDJ-400 controller into a standalone controller. So I took that skin and customized my way.

It's a bit tricky to work with the sizes here if you are not familiar with HTML and CSS.

The Mixxx skin works with XMLObjects and QSS which is similar to CSS code.

The first thing is to declare that we will use only 1 deck in the skin.xml file:

```
XML
<attribute config_key="[Master],num_decks">1</attribute>
```

We have also to declare the different *Tabs* the skin will have, so in this case I copied the CDJ3000 tabs:

- 1: Overview (waverform, main deck view)
- 2: Library
- 3: Hot Cue (to be able to trigger hot cues from the screen - like the XDJ 1000 MK2)
- 4: Beat Loop
- 5: Beat Jump
- 6: Key Shift
- 7: Stems

NOTE: Stems feature is available from Mixxx 2.6 which is the one I used. **To work with Stems you must analyze them before using a third party app. At this moment, Mixxx can't analyze it.**

The deck.xml file contains the bottom part of the deck (quantize button, time, overview waveform, pitch, tempo, key, and also **the button mode with a custom TEXT**:

```
XML
<Label>
    <ObjectName>Mode1</ObjectName>
    <Size>65f,40f</Size>
    <Text>HOT CUE 1</Text>
    <Connection>

    <ConfigKey>[Channel2],filterLowKill</ConfigKey>
        <BindProperty>visible</BindProperty>
    </Connection>
</Label>

    <Label>
    <ObjectName>Mode2</ObjectName>
    <Size>65f,40f</Size>
    <Text>HOT CUE 2</Text>
    <Connection>

    <ConfigKey>[Channel2],filterMidKill</ConfigKey>
        <BindProperty>visible</BindProperty>
    </Connection>
</Label>

    <Label>
    <ObjectName>Mode3</ObjectName>
    <Size>65f,40f</Size>
    <Text>HOT CUE 3</Text>
    <Connection>

    <ConfigKey>[Channel2],filterHighKill</ConfigKey>
        <BindProperty>visible</BindProperty>
    </Connection>
</Label>

    <Label>
    <ObjectName>Mode4</ObjectName>
    <Size>65f,40f</Size>
    <Text>ROLL</Text>
    <Connection>

    <ConfigKey>[Channel3],filterLowKill</ConfigKey>
```

```
<BindProperty>visible</BindProperty>
```

Apart from all XML files, I modified the original QSS file, so I wanted to put the same colors, font type, weight and borders like in any other Pioneer/AlphaTheta deck.



I also modified the startup image and progress bar with the project's logo:



The only thing I could not change is the library layout. According to the Mixxx community, it's included in the build, so it should be modified in the main C++ files and then re-build.



Phase 5: Linux System

In this stage, we just need to install the operating system, install Mixxx and then transfer both Skin and MIDI mapping through SCP / SSH. Of course the mapping and skin will work in any other environment such as Windows or MacOS.

For this project I used the Raspberry Pi OS that you can download (or use their app) from the Raspberry site. This OS is based on Linux.

I used the Pi OS Lite for this project, and I wanted to avoid loading the Desktop and to have additional apps and services.

When following the install instructions, you have to configure your WiFi, give the device a host name and also a username and password to access it. Then you can enable SSH.

So all the rest is made using any other computer connected to the same network, through the terminal using SSH.

Now we need to download Mixxx, configure, build and install. [You should follow the Mixxx website to build it for your Linux system.](#)

Depending on your system and OS, it might require installing some other dependencies other than the ones included with the Mixxx files.

Now that we have Mixxx installed we need to configure our operating system to run Mixxx on boot. Apart from that, we want to hide all startup messages, warnings and the splash screen.

Depending on the OS Linux system, it can be a bit different, so try and experiment with the different options.

To start Mixxx, I'm using openbox, a lightweight window manager:

```
Shell
sudo apt install openbox
```

It might require some dependencies, install them all.

And then we have to include this to the boot using a .xinitrc file:

```
Shell
nano ~/.xinitrc
```

Add the following (replace xdj100sx for the current user) (also use the Mixxx installation directory)

```
Shell
#!/bin/bash

# Start the window manager
openbox &

# Run Mixxx
exec /home/xdj100sx/mixxx/build/mixxx
```

Save and exit.

Now add run permissions to the file:

```
Shell
chmod +x ~/.xinitrc
```

Now we want to hide our mouse cursor, as we want to have the maximum user experience as any other standalone deck.

Create bash_profile file:

```
Shell
nano ~/.bash_profile
```

Add the following:

```
Shell
if [ -z "$DISPLAY" ] && [ "$(tty)" = "/dev/tty1" ]; then
    startx -- -keeppty > ~/.xorg.log 2>&1 -nocursor
fi
```

Save and exit.

Send your files using scp from a local machine:

```
Shell
scp -r /path/to/local/dir user@remotehost:/path/to/remote/dir
```

Linux does not have all the fonts by default, so we need to install additional font types in order to have the skin to look like it was designed. I've found a free version of Helvetica Neue was the best. Also, we need some additional fonts to show arrows and icons like Beatjump left/right icons.

Create this folder:

```
Shell
mkdir -p ~/.local/share/fonts
```

Move your downloaded HelveticaNeue font to the previous folder we created:

```
Shell
mv ~/Downloads/HelveticaNeue.ttc ~/.local/share/fonts/
```

Install the Noto fonts:

```
Shell
sudo apt install fonts-noto-core
```

Create this folder and edit the file:

```
Shell
mkdir ~/.config/fontconfig/
nano ~/.config/fontconfig/fonts.conf
```

Add the following:


```
XML
<?xml version="1.0"?>
<!DOCTYPE fontconfig SYSTEM "fonts.dtd">
<fontconfig>
  <match target="pattern">
    <test name="family">
      <string>Helvetica Neue</string>
    </test>
    <edit name="family" mode="prepend">
      <string>Noto Sans</string>
    </edit>
  </match>
</fontconfig>
```

Update the system fonts:

```
Shell
fc-cache -fv
```

USB Automount

This OS version does not have an “automount” feature, so everytime we connect a USB stick it won’t mount by default. The *USBmount* does this job easily:

```
Shell
git clone https://github.com/rbrito/usbmount.git
cd usbmount
sudo apt install build-essential debhelper fakeroot
dpkg-buildpackage -us -uc -b
cd ..
sudo apt install ./usbmount_0.0.24_all.deb
reboot
```

Add UTF8 charset to usbmount so, any metadata that contain non ASCII symbols in the tracks coming from a Rekordbox USB can be read and loaded properly:

```
Shell
nano /etc/usbmount/usbmount.conf
```

Find the line `FS_MOUNTOPTIONS=""` and add the following:

Shell

```
FS_MOUNTOPTIONS="-fstype=vfat,icharset=utf8,gid=plugdev,dmask=0000,fmask=0000  
-fstype=exfat,icharset=utf8,gid=plugdev,dmask=0000,fmask=0000"
```

You would also like to add ExFat compatibility, so find the `FILESYSTEMS="..."` line and include `exfat` into it;

Shell

```
FILESYSTEMS="vfat exfat ext2 ext3 ext4 hfsplus"
```

Then install Exfat Driver:

Shell

```
sudo apt install exfat-fuse exfatprogs
```

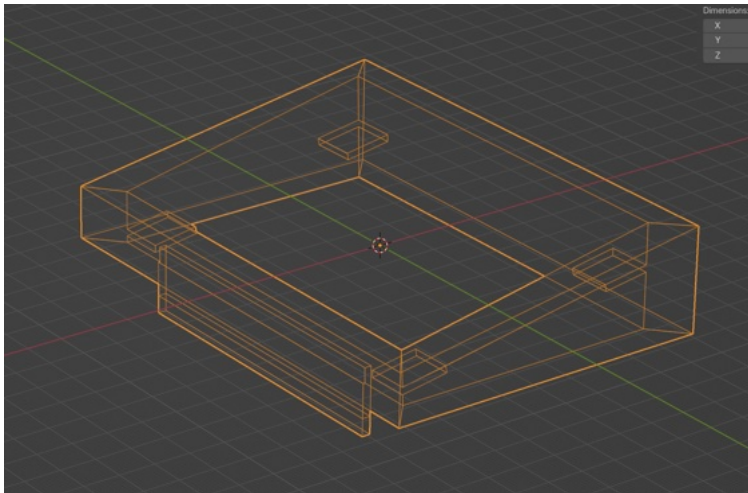
Phase 6: Physical skin

To finish this project completely, I wanted to attach the screen to the device.

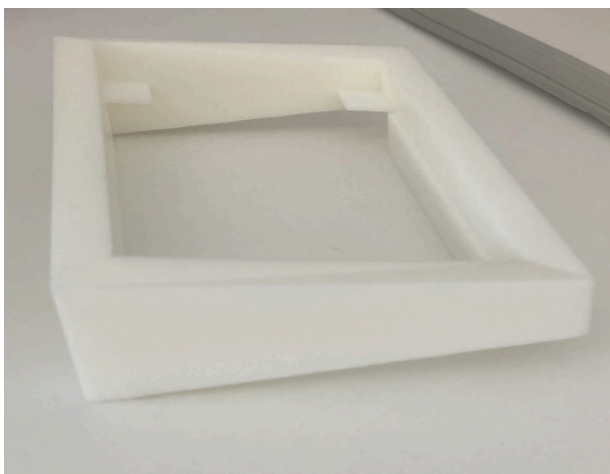
First thing, we need to **cut the original screen hole** to fit the size of our new screen:



Now we need to make a 3D model to fit the screen with the screw holes. If you are not used to it, it may take a while. I used Blender to make the model which is also an open source application of 3D modeling.



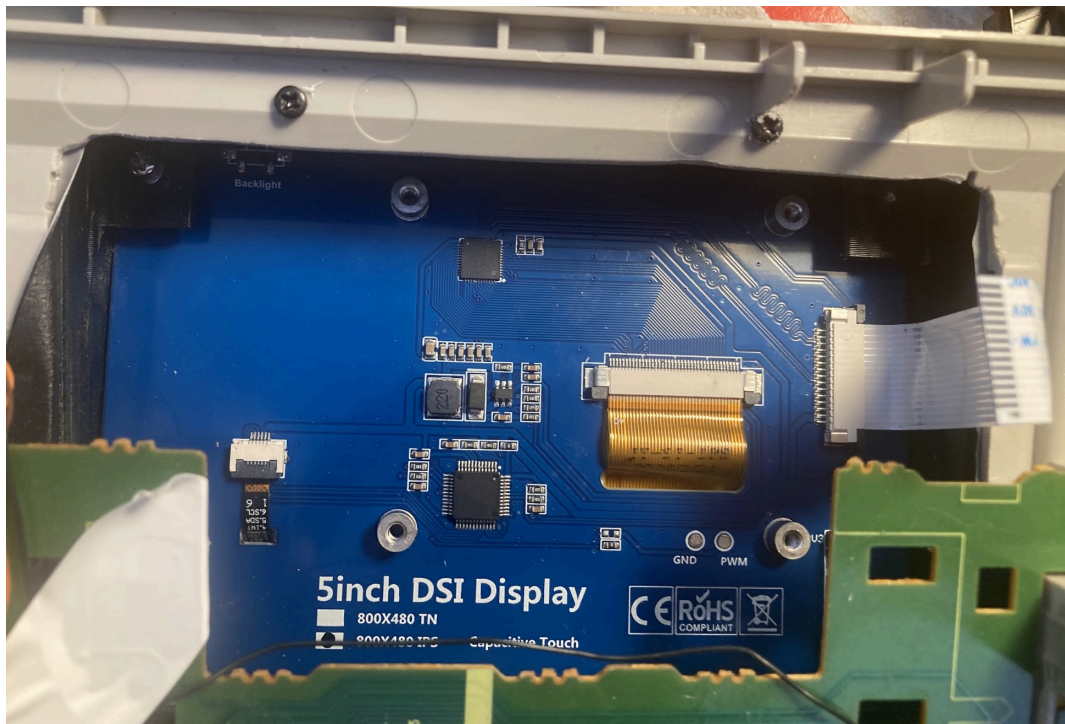
Then I printed a prototype to test. I had to make some adjustments as the screen needed 2 mm to fit in.



The final result was almost perfect, but I still needed an additional millimeter on the bottom part of the screen. I used a dremel tool to polish that part until the screen fit on it.



I could use the original screws to fix the screen into it and also the original CDJ screws to fit the 3D case into the original case:



Once this was done, I wanted to create an original vinyl printed adhesive to put it in the original case. I've found some companies that make DJ skins so I could download a template to edit and make my own, adding the custom buttons and custom functions.



And so looks that good:



Phase 7: Assemble everything

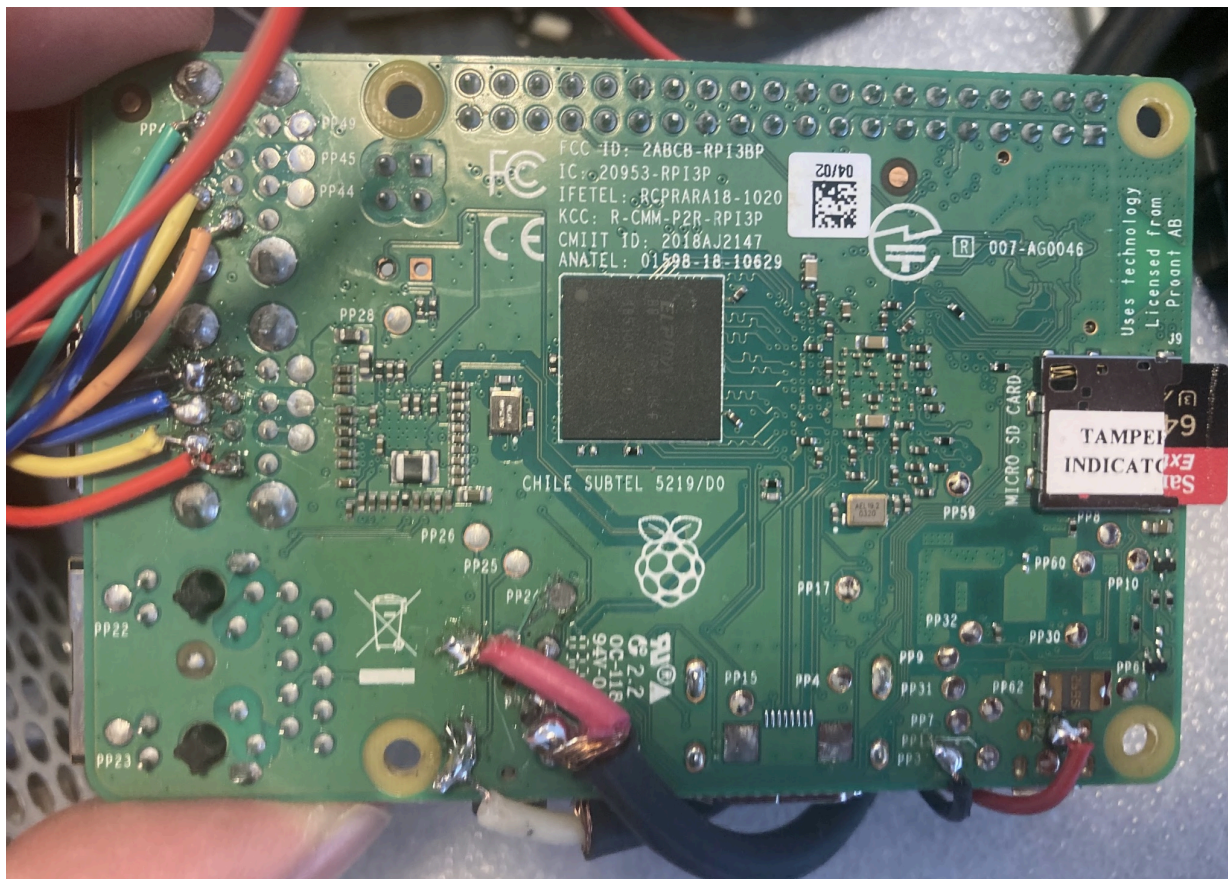
So we are in the final stage: assembling.

We need to:

- Connect Pi to the power supply
- Connect the Arduino Board to the Pi.
- Connect the external USB connector to the Pi.
- Connect the RCA connector to the Pi audio output.

As the 3D case I designed does not have space to include the USB and audio connectors, we need to solder all cables into the Pi directly.

So basically we have to look at the Pi datasheet or use the multimeter to be sure what the connectors are.



Both USB on the left. Audio on the bottom left, and power on the bottom right.

We need the wires to be as long as we can assemble with no issues. I used some terminal connectors to join hardware connections to external connectors.



The power supply is attached to the original case with some silicone.

Before adding all the screws, it's better to test it out, buttons, audio, connectors. If it's all right, then we can put all the screws.

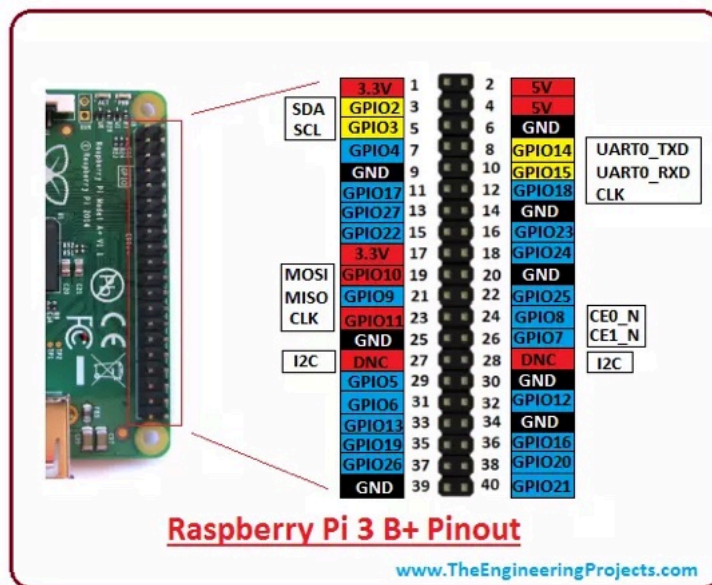


Phase 8: Improvements

8.1 Power on/off

Instead of sending a system shutoff, it first closes Mixxx so Mixxx saves all preferences, column layout, etc. Once Mixxx is closed, it powers off the system.

This Raspberry Pi 3B+ does not have any switch or button to turn it on/off. Instead, we can attach the original CDJ switch to the GPIO pins. In our case, following the datasheet, we need to use the GPIO3 and the ground. In this Pi, they correspond to PIN 5 and PIN 6:



Also, we need to close Mixxx properly so during the Mixxx exit procedure, it saves its preferences, column layout, etc.

Find the script in the GitHub repository, under /SCRIPTS. If you do not want to install the pre-made script, follow below:

Create a new script in the user folder:

```
Shell
nano shutdown_script.py
```

Copy and paste this code or adapt to your PIN.

IMPORTANT: Raspberry Pi will shut down automatically if the Pins 5 & 6 (GPIO 3) are not shorted. Be careful here otherwise you will have to make a workaround to disable it or reinstall the whole system

Python

```
#!/usr/bin/env python3
import os
import time
from gpiozero import Button

# Set the GPIO Pin - GPIO 3 = Pin 5.
pull_up=True -> shorted

switch = Button(3, pull_up=True)

try:
    while True:
        if not switch.ispressed:
            # Turn screen to black
            os.system("vcgencmd display_power 0")
            # Close Mixxx properly
            os.system("sudo -u xdj100sx env DISPLAY=:0 XAUTHORITY=/home/xdj100sx/.Xauthority
wmctrl -c Mixxx")
            # Wait until Mixxx is closed (max 20 sec)
            for i in range(20):
                if os.system("pgrep -x mixxx > /dev/null") != 0:
                    break
            time.sleep(1)
            # Black screen + shutdown
            os.system("poweroff")
            break
            time.sleep(1)
except Exception as e:
    print(f"Error: {e}")
```

Then create the service:

Shell

```
sudo nano /etc/systemd/system/shutdown-helper.service
```

Add the following:

```
Shell

[Unit]

Description=Script close Mixxx + Shutdown
After=multi-user.target

[Service]
Type=simple
ExecStart=/usr/bin/python3 /home/xdj100sx/shutdown_script.py
Restart=always
User=root
Environment=DISPLAY=:0

[Install]
WantedBy=graphical.target
```

Enable the service:

```
Shell

sudo systemctl daemon-reload
sudo systemctl start shutdown-helper.service
sudo systemctl enable shutdown-helper.service
```

8.2 USB Eject

A new service that is monitoring the "BACK" button. If you hold this button for 3 seconds, the script will unmount/eject the USB so you can unplug it properly. Note that Mixxx still keeps in cache your playlists and tracklist, but you won't be able to load any track once ejected unless you plug it back in.

We could make this through Arduino but we can just use python and rtmidi which can listen to a specific MIDI message and translate to a system action.

Requirements: python, mido, rtmidi

Shell

```
sudo apt update && sudo apt install python3-pip -y
python3 -m venv ~/venv-xdj
~/venv-xdj/bin/pip install mido python-rtmidi
```

Find the script in the GitHub repository, under /SCRIPTS. If you do not want to install the pre-made script, follow below:

Create the script:

Shell

```
sudo nano /usr/local/bin/xdj-eject.py
```

Add the following:

NOTE: modify the NOTE_BACK HEX number with the one you want to use to make the eject function. You may also want to modify the time (currently 3 seconds).

Python

```
#!/usr/bin/env python3
"""
XDJ-100SX USB Safe Eject Service
Listens for BACK held 3 seconds on the controller to safely eject the USB
drive.
Run as a systemd service alongside Mixxx.
"""

import mido
import subprocess
```

```
import time
import threading
import sys

# USB block device to eject - run `lsblk` to find the correct one
USB_DEVICE = "/dev/sda1"

# MIDI note matching the XML mapping
NOTE_BACK = 0x3F # BACK button (note_on 0x90, midino 0x3F)

# Seconds BACK must be held before ejecting
HOLD_SECONDS = 3.0

eject_timer = None

def do_eject():
    print("[eject] Syncing...", flush=True)
    subprocess.run(["sync"])

    print("[eject] Unmounting USB...", flush=True)
    r1 = subprocess.run(
        ["sudo", "umount", "-l", "/media/usb0"],
        capture_output=True, text=True
    )
    if r1.returncode != 0:
        print("[eject] Unmount error:", r1.stderr.strip(), flush=True)
        return

    print("[eject] Waiting for device to release...", flush=True)
    time.sleep(3)

    print("[eject] Powering off...", flush=True)
    r2 = subprocess.run(
        ["sudo", "udisksctl", "power-off", "-b", "/dev/sda"],
        capture_output=True, text=True
    )
    if r2.returncode == 0:
        print("[eject] Done - safe to unplug.", flush=True)
    else:
        print("[eject] Error:", r2.stderr.strip(), flush=True)

def on_message(msg):
    global eject_timer

    if msg.type not in ("note_on", "note_off"):
        return
```



```

velocity = getattr(msg, "velocity", 0)
pressed = (msg.type == "note_on" and velocity == 127)

# BACK held → start countdown
if msg.note == NOTE_BACK and pressed:
    print(f"[eject] BACK held - hold {HOLD_SECONDS}s to eject...",
flush=True)
    eject_timer = threading.Timer(HOLD_SECONDS, do_eject)
    eject_timer.start()

# BACK released before countdown ends → cancel
elif msg.note == NOTE_BACK and not pressed:
    if eject_timer is not None:
        eject_timer.cancel()
        eject_timer = None
    print("[eject] Cancelled.", flush=True)

def find_port(keyword="XDJ"):
    """Return the first input port whose name contains keyword."""
    names = mido.get_input_names()
    for name in names:
        if keyword.upper() in name.upper():
            return name
    return None

def main():
    port_name = find_port("XDJ")

    if port_name is None:
        # List available ports to help with debugging
        print("[eject] XDJ port not found. Available ports:", flush=True)
        for p in mido.get_input_names():
            print("  -", p, flush=True)
        sys.exit(1)

    print(f"[eject] Listening on: {port_name}", flush=True)
    print(f"[eject] Hold BACK for {HOLD_SECONDS}s to eject {USB_DEVICE}",
flush=True)

    with mido.open_input(port_name) as port:
        for msg in port:
            on_message(msg)

if __name__ == "__main__":

```

```
main()
```

Create the service:

```
Shell
sudo nano /etc/systemd/system/xdj-eject.service
```

Add the following:

NOTE: replace with your current system user (here it's xdj100sx)

```
Shell
[Unit]
Description=XDJ USB Eject Service
After=sound.target

[Service]
ExecStart=/home/xdj100sx/venv-xdj/bin/python3 /usr/local/bin/xdj-eject.py
Restart=always
User=xdj100sx

[Install]
WantedBy=multi-user.target
```

Enable the service:

```
Shell
sudo systemctl daemon-reload
sudo systemctl restart xdj-eject
sudo systemctl status xdj-eject
```

8.3 USB Network Share

You can now load music from the other's xdj100sx USB. You only need both devices to be connected to the same network. It works with both WiFi and Ethernet. In Mixxx you will see, under Rekordbox, "USB1" which is the one connected to the second device. The Pi3B+ WiFi is not so fast so you may expect a bit of lag when scrolling the network usb drive, but it loads properly.

How it works?

- Plug in the USB to the XDJ100SX device 1.
- Use the XDJ100SX device 1 as usual.
- On the XDJ100SX device 2, under the Mixxx library, find the network USB under Rekordbox - usb1.
- See your playlists, scroll and load the track.

Find the script in the GitHub repository, under /SCRIPTS. You need first to install usbmount.

```
Shell
sudo apt install usbmount
```

Install script:

```
Shell
sudo bash SCRIPTS/usb-share/install.sh
```

The script does the following:

- Creates a service that listens new USB connections and share automatically over the network using NFS
- Creates a second service that watch the network to find a shared USB and mounts it.

Using the image

If you have a **Raspberry Pi 3B+ model** you can use the image file attached in the GitHub repository and restore it to your SD card using Raspberry Pi Imager application from any computer. Choose the Raspberry 3B+ and select 'Use a custom image'. Then select the IMG.GZ file.

IMPORTANT: some users said the image works with Pi 4 although it's not been tested.

XDJ100SX image user: xdj100sx

XDJ100SX image password: xdj100sx

Things to do once using the image:

1. Connect a USB keyboard to the USB port
2. Hold Control + Alt + F2 to enter the console
3. Put the user xdj100sx and password xdj100sx
4. Connect to your WiFi:

```
Shell
sudo raspi-config
```

You will enter to the Raspberry Pi configuration. Find the Wireless settings, add your SSID (WiFi name) and then the WiFi password manually.

Check XDJ100SX IP address:

From the console (Hold control + Alt + F2 if you are on Mixxx):

```
Shell
ifconfig
```

It will show you all Pi network devices, find the wlan0 or similar and you should see something similar to ip address XXX.XXX.X.XXX (for example 192.168.1.132)

SSH Connection:

From a MacOS / Windows / Linux computer connected to the same network, open the command terminal and type:

```
Shell
ssh xdj100sx@the-xdj-ip-address
```

You can now operate, change Raspi settings, add scripts, using your computer.

Contact: info@marcmonka.com
[Instagram](#)
[Facebook](#)



Join the community:
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